Software Requirements Specification (SRS)

Project: Academic E-Commerce & Identity Management Portal (Demo Specification)

1. User Authentication & Session Management

REQ-001: The system shall allow registered users to log in using their credentials.

REQ-002: The system should be very fast.

REQ-003: The system shall respond within 2 seconds under standard operational workload.

REQ-004: Users can reset their password.

REQ-005: Users can change their forgotten password.

2. Analytics & Reporting

REQ-006: The system shall generate reports.

REQ-007: The administrator shall generate reports using PostgreSQL and export them to PDF format.

REQ-008: The platform shall provide interactive sales summary charts for quarterly reviews.

3. Scalability & System Performance

REQ-009: The system shall support at least 10,000 concurrent users.

REQ-010: The system should support a large number of users.

REQ-011: Database transactions shall maintain an average commit latency of under 50 milliseconds.

4. Security & Cryptographic Policies

REQ-012: Passwords must contain at least 8 characters.

REQ-013: Passwords must contain a maximum of 6 characters.

REQ-014: All sensitive customer payment records must be encrypted at rest using AES-256 cipher standard.

REQ-015: The system shall enforce role-based access control (RBAC) for all administrative endpoints.

5. Order Tracking & Customer Service

REQ-016: The system shall allow customers to track orders and view shipment status.

REQ-017: Customers shall receive real-time email notifications when their order status changes.

REQ-018: The application shall validate credit card numbers using the Luhn algorithm before submission.